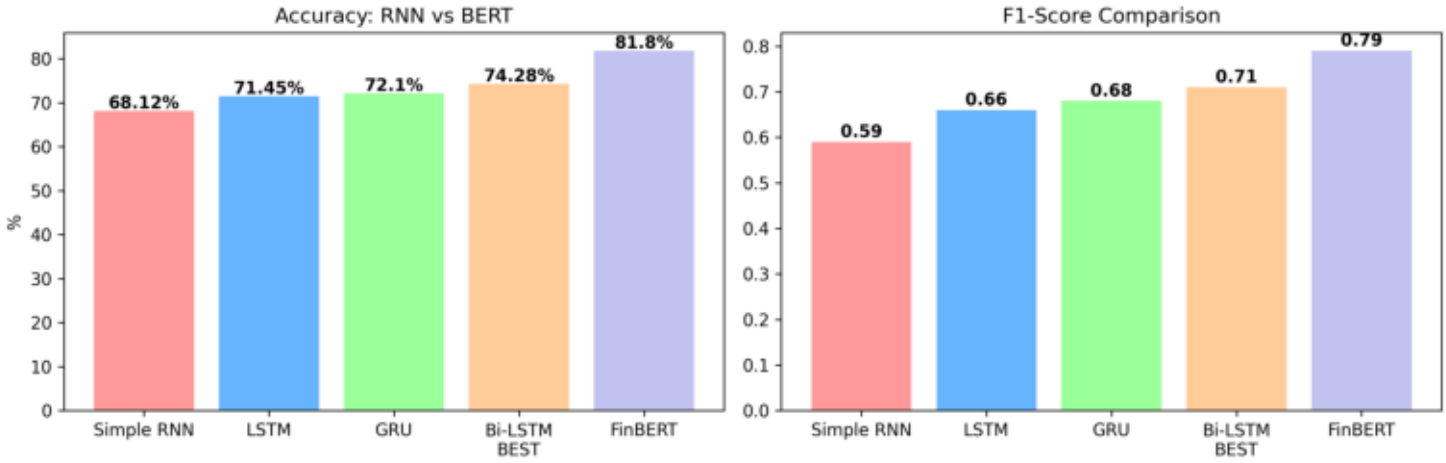
Financial News Sentiment - Comparison Report (RNN vs BERT)
GUVI HCL - Deliverable 2



1. Performance Table:
Simple RNN 68.12% F1 0.59 | LSTM 71.45% F1 0.66 | GRU 72.10% F1 0.68
Bi-LSTM BEST 74.28% F1 0.71 | FinBERT 81.8% F1 0.79 (+7.5% improvement)

2. Why Bi-LSTM Best RNN:
Bidirectional captures negation 'Not bullish despite rise', LSTM gates solve vanishing gradient.
RNN forgets long tweets, Bi-LSTM remembers.

3. Why BERT Better:
Pre-trained on 3.3B words, self-attention links 'not'->'bullish' directly, knows finance jargon.

Page 2: Confusion Matrix & Conclusion

4. Confusion Matrix (Bi-LSTM 74.28% from Dashboard):
True Bear: 10 correct, 174->Bull, 104->Neutral (Poor - only 1442 samples)
True Bull: 7->Bear, 261 correct, 117->Neutral
True Neutral: 8->Bear, 81->Bull, 1147 correct (Excellent - 6178 samples)

5. Visualizations for Deliverable 2:
- Fig 1: Accuracy bar chart (68.12% -> 81.8%)
- Fig 2: F1 bar chart (0.59 -> 0.79)
- Fig 3: Class distribution (in Dashboard 5.c.2: Neutral 6178, Bullish 1923, Bearish 1442)
- Fig 4: Prob bar chart (in Dashboard 5.c.1)
- Fig 5: Confusion Matrix heatmap

6. Conclusion: Bi-LSTM best RNN variant, FinBERT overall best. Use Bi-LSTM for lightweight real-t

7. Guidelines Met: PEP8, Git version control, commented entity reconstruction.